# Algebraic Number Theory: Study Notes

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# Note

These notes provide a gradual introduction to algebraic number theory, beginning with the necessary algebraic foundations before developing the theory of number field. The exposition emphasizes clear definitions, rigorous proofs, and intuition, with the aim of building a solid understanding of the subject. This is a living document that will be continually revised, expanded, and refined as my study of algebraic number theory progresses, with new material, examples, exercises, and references incorporated over time.

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# Chapter 1

# Introduction to Commutative Algebra

## 1.1 Rings and Ideals

Commutative algebra begins with the study of rings and their ideals. These objects form the basic language for much of modern algebra, algebraic geometry, and number theory. In this chapter we introduce the fundamental definitions and properties that will be used throughout the text.

### Rings and ring homomorphisms

A *ring* $R$ is a set equipped with two binary operations, addition and multiplication, such that:

1. $(R, +)$ is an abelian group with identity element 0;
2. multiplication is associative and distributive over addition;
3. the ring is commutative: $xy = yx$ for all $x, y \in R$;
4. $R$ has an identity element 1 such that $1x = x1 = x$ for all $x \in R$.

Unless stated otherwise, every ring in this text is assumed to be commutative and to have identity. The case $1 = 0$ is allowed; then the ring consists of a single element and is called the *zero ring*.

A *ring homomorphism* is a map $f : R \to S$ such that

$$f(x + y) = f(x) + f(y), \qquad f(xy) = f(x)f(y), \qquad f(1) = 1.$$

A subset $A \subseteq R$ is called a *subring* if it is closed under addition and multiplication and contains 1.

### Ideals and quotient rings

An *ideal* $I$ of a ring $R$ is an additive subgroup of $R$ such that

$$RI \subseteq I.$$

Equivalently, for every $x \in R$ and $y \in I$ we have $xy \in I$.

If $I$ is an ideal of $R$, then the quotient group $R/I$ becomes a ring under the rule

$$(x + I)(y + I) = xy + I.$$

This ring is called the *quotient ring* or *residue class ring* of $R$ modulo $I$.

There is a natural surjective ring homomorphism

$$\phi : R \to R/I, \qquad x \mapsto x + I,$$

whose kernel is precisely $I$.

### Prime and maximal ideals

An ideal $P \subsetneq R$ is called *prime* if

$$xy \in P \implies x \in P \text{ or } y \in P.$$

An ideal $M \subsetneq R$ is called *maximal* if there is no ideal $J$ such that

$$M \subsetneq J \subsetneq R.$$

These notions are closely related to familiar algebraic structures:

- $P$ is prime if and only if $R/P$ is an integral domain;
- $M$ is maximal if and only if $R/M$ is a field.

Every maximal ideal is prime. Also, a ring $R \neq 0$ has at least one maximal ideal.

### Zero-divisors, nilpotent elements, and units

An element $x \in R$ is a *zero-divisor* if there exists $y \neq 0$ in $R$ such that $xy = 0$. A ring with no zero-divisors is called an *integral domain*.

An element $x \in R$ is *nilpotent* if $x^n = 0$ for some $n > 0$.

An element $x \in R$ is a *unit* if there exists $y \in R$ such that $xy = 1$. The set of all units of $R$ forms a multiplicative group.

### Operations on ideals

If $I$ and $J$ are ideals of $R$, then their *sum*, *intersection*, and *product* are defined by

$$I + J = \{x + y : x \in I,\ y \in J\},$$

$$I \cap J,$$

and

$$IJ = \left\{ \sum_k x_k y_k : x_k \in I,\ y_k \in J \right\}.$$

These operations satisfy the usual commutative and associative laws, and the product distributes over sum:

$$I(J + K) = IJ + IK.$$

Two ideals $I$ and $J$ are said to be *coprime* if

$$I + J = R.$$

In this case,

$$I \cap J = IJ.$$

## Radicals

The *radical* of an ideal $I$ is

$$\sqrt{I} = \{x \in R : x^n \in I \text{ for some } n > 0\}.$$

It is again an ideal of $R$. The radical of the zero ideal is called the *nilradical* and is the set of all nilpotent elements of $R$.

The *Jacobson radical* of $R$ is the intersection of all maximal ideals of $R$.

## Extension and contraction

If $f : R \to S$ is a ring homomorphism and $I$ is an ideal of $R$, the *extension* of $I$ to $S$ is the ideal generated by $f(I)$ in $S$.

If $J$ is an ideal of $S$, the *contraction* of $J$ to $R$ is

$$J^c = f^{-1}(J).$$

These two operations play an important role in the study of ideals under ring homomorphisms.

The theory developed in this chapter will be used repeatedly in later chapters. Prime and maximal ideals, quotient rings, radicals, and ideal operations form the core of commutative algebra.